



Wireless BABY MONITOR

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This wireless baby monitor circuit using FM is designed to operate at a frequency of 100MHz. Its range of transmission is more than 100 metres with a 75cm wire antenna. The circuit is fully transistorised and so even a beginner can easily assemble it on a veroboard or the PCB whose pattern is given below for convenience.

Circuit and working

Circuit diagram of the wireless baby monitor circuit is shown in Fig. 1. It is built around a condenser microphone (MIC1), transistors BC549 (T1 and T3) and BC557 (T2), along

with a few other components.

T1 and T2 provide high-gain audio to the VHF oscillator wired around T3. BC549 can oscillate well in VHF range. Frequency-modulated (FM) signals are transmitted through antenna ANT.1.

By adjusting trimmer capacitor VC1, frequency can be set within 88MHz -108MHz band. Using a good FM radio (analogue type) receiver, transmitted signals can be heard. Do not use a mobile phone as receiver.

This unit with 9-volt battery can be kept near a baby's bed. Any sound generated by the baby including a cry can be monitored wire-

PARTS LIST

Semiconductors:

- T1, T3 - BC549 npn transistor
- T2 - BC557 pnp transistor
- Resistors (all 1/4-watt, 45% carbon):
- R1, R5-R7 - 10-kilo-ohm
- R2 - 100-kilo-ohm
- R3 - 150-kilo-ohm
- R4 - 12-kilo-ohm
- R8 - 680-ohm
- R9 - 22-kilo-ohm
- R10 - 2.2-kilo-ohm
- R11 - 68-ohm

Capacitors:

- C1 - 100µF, 25V electrolytic
- C2, C4 - 0.1µF ceramic disk
- C3, C5 - 10µF, 25V electrolytic
- C6 - 1nF ceramic disk
- C7 - 10pF ceramic disk
- C8 - 100pF ceramic disk
- C9 - 0.01µF ceramic disk

Miscellaneous:

- CON1 - 2-pin terminal connector
- ANT.1 - 75-cm-long wire
- MIC1 - COND. MIC
- L1 - 0.1µH inductor
- L1 - 9V battery
- FM radio (analogue type)

lessly from another room. Readily available 1-micro Henry inductor can be used as L1. It can also be home brewed as shown in Fig. 2.

Construction and testing

An actual-size, single-side PCB pattern of the wireless baby monitor circuit is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on a PCB, enclose it in a plastic case and keep it near the baby's bed.

The circuit works well off a 9V battery.

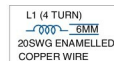


Fig. 2: Inductor details

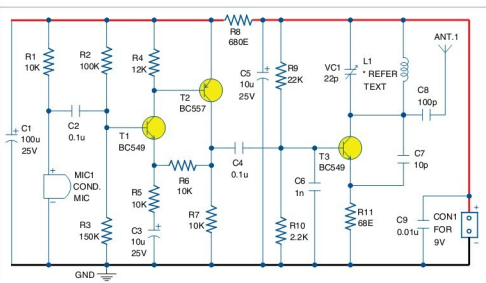


Fig. 1: Circuit of the wireless baby monitor

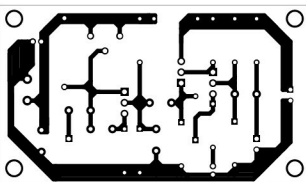


Fig. 3: Actual-size PCB pattern of the wireless baby monitor

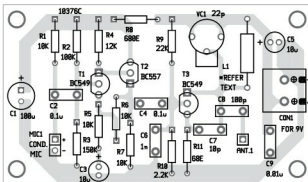


Fig. 4: Component layout of the PCB



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